

## NUERBURG BARWEILER, Germany

WMO: 105060

Lat: 50.3603N

Lon: 6.8697E

Elev: 1602

StdP: 13.87

Time Zone: 1.00 (EUC)

Period: 95-13

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 14.6       | 18.2 | 7.4                           | 8.5 | 19.2 | 12.2 | 10.9 | 22.4 | 25.7                  | 42.9 | 22.8 | 42.5 | 6.5                   | 100  | 0.578 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 15.1                   | 81.6            | 64.2 | 77.9 | 63.0 | 74.7 | 61.8 | 66.4                | 77.1 | 64.8 | 74.3 | 63.3 | 71.8 | 7.7                  | 220  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 62.8                            | 90.9 | 69.6 | 61.3 | 85.9 | 68.4 | 59.9 | 81.7 | 67.0 | 31.9          | 77.4 | 30.7 | 74.5 | 29.5 | 71.7 | 74.5              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |     |                    |     | n-Year Return Period Values of Extreme Temperature |     |            |     |            |      |            |      |      |     |
|-------------------|------|------|----------------------------|-----|--------------------|-----|----------------------------------------------------|-----|------------|-----|------------|------|------------|------|------|-----|
|                   |      |      | Mean                       |     | Standard Deviation |     | n=5 years                                          |     | n=10 years |     | n=20 years |      | n=50 years |      |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max | Min                | Max | Min                                                | Max | Min        | Max | Min        | Max  | Min        | Max  |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f) | (g)                | (h) | (i)                                                | (j) | (k)        | (l) | (m)        | (n)  | (o)        | (p)  |      |     |
| (4) 20.9          | 18.0 | 15.8 | DB                         | 9.1 | 87.8               | 5.9 | 3.9                                                | 4.9 | 90.6       | 1.4 | 92.8       | -1.9 | 95.0       | -6.2 | 97.8 | (4) |
| (5)               |      |      | WB                         | 8.3 | 69.0               | 5.8 | 1.3                                                | 4.1 | 69.9       | 0.7 | 70.7       | -2.6 | 71.4       | -6.8 | 72.3 | (5) |

## Monthly Climatic Design Conditions

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 47.2   | 32.7  | 34.1  | 39.6  | 46.3  | 53.3  | 58.4  | 62.0  | 61.8  | 55.5  | 48.5  | 40.1  | 33.5 |
|                                                                               | DBStd       | 12.67  | 7.90  | 7.94  | 7.58  | 7.46  | 6.97  | 6.57  | 6.14  | 5.99  | 5.86  | 6.83  | 6.54  | 7.39 |
|                                                                               | HDD50       | 2459   | 537   | 445   | 330   | 159   | 46    | 6     | 0     | 0     | 13    | 110   | 302   | 511  |
|                                                                               | HDD65       | 6596   | 1002  | 865   | 788   | 562   | 365   | 216   | 136   | 136   | 291   | 511   | 748   | 976  |
|                                                                               | CDD50       | 1446   | 0     | 0     | 7     | 47    | 150   | 257   | 371   | 367   | 178   | 65    | 4     | 0    |
|                                                                               | CDD65       | 107    | 0     | 0     | 0     | 0     | 3     | 18    | 42    | 38    | 6     | 0     | 0     | 0    |
|                                                                               | CDH74       | 862    | 0     | 0     | 0     | 1     | 32    | 145   | 315   | 331   | 37    | 1     | 0     | 0    |
|                                                                               | CDH80       | 175    | 0     | 0     | 0     | 0     | 2     | 20    | 68    | 84    | 1     | 0     | 0     | 0    |
| Wind                                                                          | WSAvg       | 7.5    | 8.1   | 8.6   | 8.1   | 8.0   | 7.2   | 6.8   | 7.2   | 6.3   | 6.8   | 7.6   | 7.5   | 7.9  |
| Precipitation                                                                 | PrecAvg     | 30.2   | 2.6   | 2.3   | 2.2   | 1.9   | 2.5   | 2.8   | 2.8   | 2.9   | 2.4   | 2.4   | 2.5   | 2.9  |
|                                                                               | PrecMax     | 36.7   | 7.2   | 5.7   | 4.4   | 4.1   | 4.7   | 5.7   | 6.6   | 6.4   | 5.6   | 5.8   | 4.1   | 7.6  |
|                                                                               | PrecMin     | 22.3   | 0.5   | 0.3   | 0.1   | 0.1   | 0.8   | 0.9   | 1.4   | 0.9   | 0.6   | 0.8   | 0.1   | 0.4  |
|                                                                               | PrecStd     | 3.7    | 1.3   | 1.2   | 1.2   | 0.9   | 1.0   | 1.1   | 1.3   | 1.6   | 1.4   | 1.1   | 0.9   | 1.5  |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 50.9  | 53.4  | 62.8  | 72.1  | 78.1  | 83.3  | 85.9  | 88.1  | 78.4  | 68.8  | 56.8  | 50.9 |
|                                                                               |             | MCWB   | 48.8  | 46.6  | 50.0  | 54.3  | 60.9  | 65.6  | 65.2  | 66.1  | 61.6  | 59.3  | 51.6  | 48.4 |
|                                                                               | 2%          | DB     | 48.5  | 49.4  | 57.6  | 67.6  | 72.6  | 78.2  | 81.5  | 81.3  | 73.4  | 63.9  | 53.3  | 48.6 |
|                                                                               |             | MCWB   | 46.5  | 44.5  | 47.8  | 53.6  | 59.2  | 63.9  | 64.2  | 64.2  | 61.5  | 57.1  | 50.1  | 46.7 |
|                                                                               | 5%          | DB     | 46.1  | 47.0  | 53.9  | 62.9  | 69.3  | 74.3  | 77.3  | 77.3  | 69.1  | 60.4  | 51.2  | 46.2 |
|                                                                               |             | MCWB   | 44.2  | 43.8  | 46.9  | 51.5  | 56.9  | 62.0  | 63.1  | 62.5  | 59.6  | 55.9  | 49.0  | 44.8 |
|                                                                               | 10%         | DB     | 43.2  | 44.7  | 50.4  | 58.6  | 65.7  | 70.5  | 73.7  | 73.4  | 65.3  | 57.6  | 48.8  | 43.7 |
|                                                                               |             | MCWB   | 41.4  | 42.3  | 45.5  | 49.4  | 55.5  | 60.4  | 61.9  | 61.6  | 58.1  | 54.2  | 46.9  | 42.4 |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 49.3  | 48.4  | 52.1  | 57.2  | 64.4  | 67.9  | 68.2  | 68.2  | 64.6  | 60.7  | 53.1  | 49.4 |
|                                                                               |             | MCDB   | 50.2  | 51.0  | 57.9  | 67.2  | 73.6  | 78.2  | 79.4  | 81.1  | 72.2  | 65.9  | 55.7  | 50.4 |
|                                                                               | 2%          | WB     | 46.7  | 46.2  | 49.9  | 54.8  | 61.2  | 65.3  | 66.3  | 65.9  | 62.4  | 58.3  | 50.8  | 47.0 |
|                                                                               |             | MCDB   | 47.9  | 48.3  | 55.2  | 64.5  | 69.7  | 74.7  | 76.8  | 77.6  | 70.4  | 62.1  | 52.7  | 48.2 |
|                                                                               | 5%          | WB     | 44.4  | 44.4  | 47.9  | 52.6  | 58.6  | 63.5  | 64.7  | 64.3  | 60.6  | 56.4  | 49.1  | 44.8 |
|                                                                               |             | MCDB   | 45.9  | 46.4  | 52.9  | 60.7  | 67.0  | 72.2  | 74.5  | 73.0  | 67.6  | 59.7  | 51.0  | 46.0 |
|                                                                               | 10%         | WB     | 41.5  | 42.4  | 46.0  | 50.6  | 56.4  | 61.4  | 63.0  | 62.6  | 58.7  | 54.5  | 47.0  | 42.3 |
|                                                                               |             | MCDB   | 42.9  | 44.4  | 49.5  | 57.8  | 63.4  | 68.9  | 71.4  | 70.8  | 64.3  | 57.5  | 48.7  | 43.5 |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 7.3   | 8.7   | 11.6  | 14.3  | 14.8  | 15.3  | 15.1  | 15.5  | 13.3  | 10.8  | 7.7   | 7.0  |
|                                                                               |             | MCDBR  | 8.9   | 11.8  | 17.5  | 20.5  | 20.7  | 21.3  | 21.1  | 22.5  | 19.2  | 14.1  | 9.9   | 7.8  |
|                                                                               |             | MCWBR  | 8.1   | 8.2   | 10.2  | 10.1  | 10.6  | 10.3  | 8.8   | 9.4   | 9.6   | 8.3   | 7.6   | 7.2  |
|                                                                               | 5% WB       | MCDBR  | 8.6   | 9.3   | 13.9  | 18.0  | 18.7  | 18.8  | 18.8  | 18.7  | 16.5  | 12.7  | 8.9   | 7.4  |
|                                                                               |             | MCWBR  | 8.4   | 7.6   | 9.1   | 9.4   | 10.3  | 10.1  | 8.8   | 9.1   | 9.0   | 8.2   | 7.3   | 7.4  |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        | 0.300  | 0.324 | 0.356 | 0.388 | 0.394 | 0.396 | 0.397 | 0.392 | 0.363 | 0.345 | 0.320 | 0.297 |      |
|                                                                               | taud        | 2.443  | 2.392 | 2.324 | 2.272 | 2.284 | 2.293 | 2.284 | 2.328 | 2.389 | 2.437 | 2.441 | 2.428 |      |
|                                                                               | Ebn at Noon | 236    | 258   | 267   | 271   | 274   | 273   | 271   | 266   | 262   | 246   | 224   | 218   |      |
|                                                                               | Edh at Noon | 19     | 26    | 33    | 39    | 40    | 40    | 40    | 36    | 30    | 24    | 19    | 17    |      |
| All-Sky Solar<br>Radiation                                                    | RadAvg      | 243    | 465   | 826   | 1313  | 1543  | 1704  | 1611  | 1385  | 1029  | 599   | 277   | 186   |      |
|                                                                               | RadStd      | 38     | 88    | 126   | 156   | 148   | 161   | 186   | 137   | 128   | 61    | 36    | 27    |      |

## Historical Trends

|                              | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only            | N/A   | N/A     | N/A    | N/A     | +1.28 | +1.40 | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (64 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page